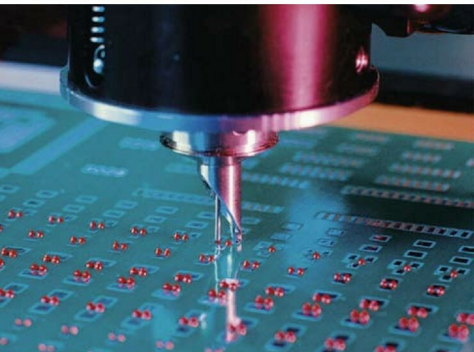


SMT Equipment

Many A Tech Trend Sighted

Economic downturn or not, the SMT equipment space is witnessing several technological trends strengthening, ranging from RoHS-ready production environments and highly flexible apparatus, to automated inspection and greater software integration



■ JANANI GOPALAKRISHNAN VIKRAM

You do not require an engineer to tell you that electronic products are getting smaller and smaller—from modems to computers. Even the electronics embedded in household gadgets like refrigerators are getting smaller. Nobody has the space to house large electronics anymore! As a result, printed circuit boards (PCBs) and the components mounted on them are getting increasingly miniaturised.

One of the factors responsible for

this miniaturisation is the surface mount technology (SMT) that was developed by IBM way back in the 1960s. Simply put, SMT is a way of making electronic circuits in which the components are mounted directly onto the surface of the PCBs, instead of the earlier through-hole construction method. SMT equipment refers to the broad range of machines used for the production of PCBs through the surface mount technique—placement equipment, screen printing equipment, automated testers, rework and reflow stations, soldering equipment, etc.

The constant evolution of SMT

equipment is a key to the increasing capability and decreasing size of electronics. Have you seen artists paint the Taj Mahal on a grain of rice? Manufacturing a PCB today requires more precision than that. Do you know that some of those artists paint not just the Taj Mahal but portraits of visitors too? And they do it while you wait next to their desk! That is how flexible and fast SMT equipment have to be too today, to print circuit boards fast enough to meet the extremely challenging time-to-market requirements of electronic manufacturing service (EMS) companies and electronics manufacturers.

Technologically classy

"There is a recent demand in the industry to produce high-precision complex boards and assemblies. At the same time, there is constant pressure to increase the quality of the goods produced, aiming for zero defects," says Pradeep Bhatnagar, vice president-operations, Rangsons Electronics, Mysore. To be able to manage this, manufacturers need to have the right SMT equipment which will enable fast, flexible and perfect production of PCBs.

"To overcome the challenges posed by international competition, Indian EMS companies, especially those in SMT production, have to be careful while selecting the latest technology, focusing on their business plans, what segment they cater to and sticking to their best potential," he adds.

Nowadays a wide range of machines are available—throughput as